

Optimal Control of a Continuum Manipulator

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Abstract—We develop appropriate reduced order model of for beam dynamics and show their use for direct collocation control of a continuum manipulator. We found control inputs that result in minimum time trajectories to certain poses. We discovered the minimum number of knot points for this particular problem. We manage to speed up the computation time significantly by supplying analytical Jacobians for the constraint matrix and the objective function. Finally we use trajectory optimization to discover which parts of the manipulator workspace are reachable.

I. INTRODUCTION

Continuum robots have promise for their capacity to work with humans and animals naturally in their own environments. This is because of their inherently compliant and back-drivable nature. These same properties that allow them to interact with humans and animals [1] in ways that are arguably more safe and natural than segmented rigid robots also makes them more difficult to control with conventional methods. Soft robots are controlled or actuated by inducing deformations in the manipulator cross section by applying fluid pressure. Consequently modeling the dynamics is non-trivial and an open area of research.

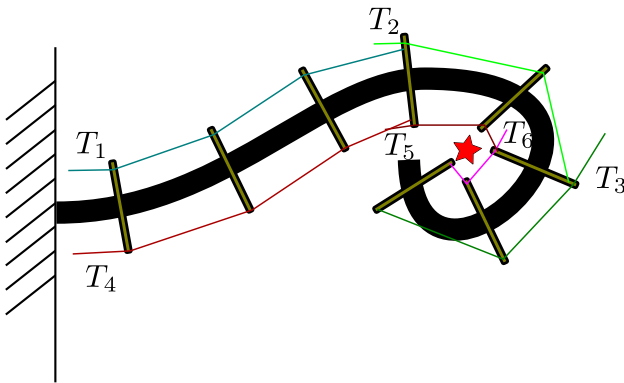


Fig. 1. Concept Image

II. A CONTINUUM BEAM AS A PLANT

In this work we would seek to control a continuum manipulator. One simple kind utilizes tendons running along the length of the manipulator on two sides these allow the imposition of a constant force. Putting more than one tendon as shown in figure 1 by tension forces denoted by T_i allows the manipulator to have various curvatures along its length.

III. BACKGROUND

As mentioned, fast continuum beam modeling for control is an ongoing area of research. For instance one approach is to model is using finite difference equations that approximate the governing equations in space and time [2]. It is difficult to include cross sectional deformations in the FD approach. Another would be to use finite element. Practically this is very slow. Alternatively, Walker [5] mentions there is a way to use Langrangian dynamics to derive dynamic equations for continuum manipulators. Marchese and Tedrake show one way to compute equations of motion (EOM) and use direct collocation for a soft actuators [3] Finally Althoff also presents a method based on modal analysis for controlling a beam [4] this method utilizes modal analysis to extract a coupled system of ODEs that govern the system.

IV. SIMULATION / MODEL

Marchese's[3] method was followed to determine dynamic equations of motion for the continuum beam. There is a commonly used transformation in continuum robotics for finding translations and rotations along a manipulator [5]. This project makes the piece-wise continuous assumption, and as such, the following equation gives the transformation:

$$S_{tip}^{base}(\theta) = \begin{pmatrix} \cos(\theta) & \sin(\theta) & \frac{L(1-\cos(\theta))}{\theta} \\ -\sin(\theta) & \cos(\theta) & \frac{L\sin(\theta)}{\theta} \end{pmatrix} \quad (1)$$

This study includes two elements with constant curvature and stiffness such that the position of these masses in a global reference is given by the transformation

$$\vec{P}_1 = S(\theta_1) \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}^T$$

$$\vec{P}_2 = S(\theta_1)S(\theta_2) \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}^T$$

$$\vec{P}_n = S(\theta_1)S(\theta_2)...S(\theta_n) \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}^T$$

Equations of motion are then developed utilizing the Euler-Lagrange method, yeilding two coupled ODEs.

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} - \frac{\partial \mathcal{L}}{\partial \theta_1} = -b\dot{\theta}_1 + \tau_1 - \tau_2$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_2} - \frac{\partial \mathcal{L}}{\partial \theta_2} = -b\dot{\theta}_2 + \tau_2$$

where

$$\mathcal{L} = Ke - (pe_{compliance} + pe_{gravity}) \text{ and } F_n = \frac{\partial Pe}{\partial \theta_n}$$

Potential energy is saved when the element flexes according to the law:

$$pe_{compliance} = \frac{1}{2}k\theta_1^2 + \frac{1}{2}k\theta_2^2$$

$$pe_{gravity} = mgP_1(z) + mgP_2(z)$$

Kinetic energy is found by differentiating the position vectors $\vec{P}_1$ and $\vec{P}_2$ wrt time to get velocity vectors $\vec{v}_1$ and $\vec{v}_2$

$$Ke = \frac{1}{2}m(\vec{v}_1^T \vec{v}_1 + \vec{v}_2^T \vec{v}_2)$$

The $\frac{\sin(x)}{x}$ in the transformation matrix created difficulty for the time marching numerical integration. I decided to deal with this difficulty by replacing $\frac{\sin(x)}{x}$ with its sixth order Taylor series equivalent. This was done before applying Euler-Lagrange to generate the EOMs.

1) *Validating the Model:* In figure 2 we demonstrate that with no control input the sum of kinetic and potential energy remain constant and the system evolves in time according to expectation. And serves as a “sanity check” on the derived Lagrangian Dynamics

Similarly, figure 3 shows results of adding damping and control effort to the system. And it behaves as expected. Note that these simulations were done in a different way from the later optimizations. for these, the EOMs were explicitly solved in the sense of

$$M\dot{x} = Cx + g$$

therefore

$$\dot{x} = M^{-1}(Cx + g)$$

this worked fine for simulation, but proved to be a problem for optimization. Details on the solution to follow.

V. OPTIMIZATION

For this study we are interested in finding the control inputs that minimize the travel time of the soft manipulator as it moves to a final specified pose. This motion is subject to actuator limitations. We approach this by first transcribing the problem as an optimization problem.

$$\begin{aligned} &\underset{x}{\text{minimize}} && t(x) \\ &\text{subject to} && f_i(x) \leq b_i, \quad i = 1, \dots, m. \end{aligned}$$

No Control No Damping, Params: $k = 2 \quad b = 0$

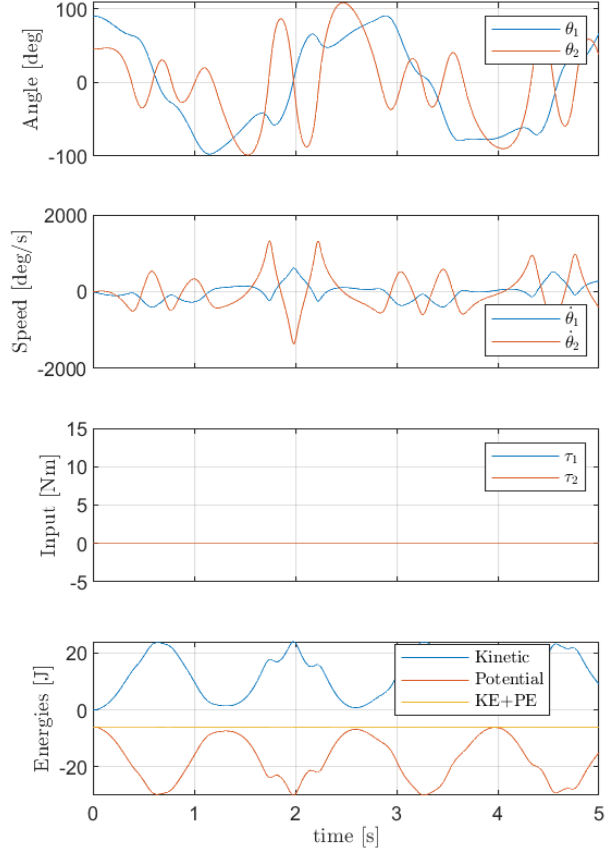


Fig. 2. Result of ODE45 integration of system with no damping and no control input. Marches forward in time conserving energy as expected.

A. Constraints

The optimization is subject to $-5 < \tau < 5$ so control effort limited.

$$\tau_{min} < \begin{pmatrix} \tau_{11} \\ \vdots \\ \tau_{1n} \\ \tau_{21} \\ \vdots \\ \tau_{2n} \end{pmatrix} < \tau_{max}$$

The desire to end at a certain pose is handled as an equality constraint. Where the final knot point states are set equal to a final desired state:

$$\begin{pmatrix} \theta_{1n} \\ \theta_{2n} \\ \dot{\theta}_{1n} \\ \dot{\theta}_{2n} \end{pmatrix} = \begin{pmatrix} \theta_{1final} \\ \theta_{2final} \\ \dot{\theta}_{1final} \\ \dot{\theta}_{2final} \end{pmatrix}$$

Defect constraints enforce the system dynamics with trapezoidal integration approximation. Note here that unlike during simulation, these dynamics are written in the

Non-Zero Inputs: $k = 2$ $b = 2$

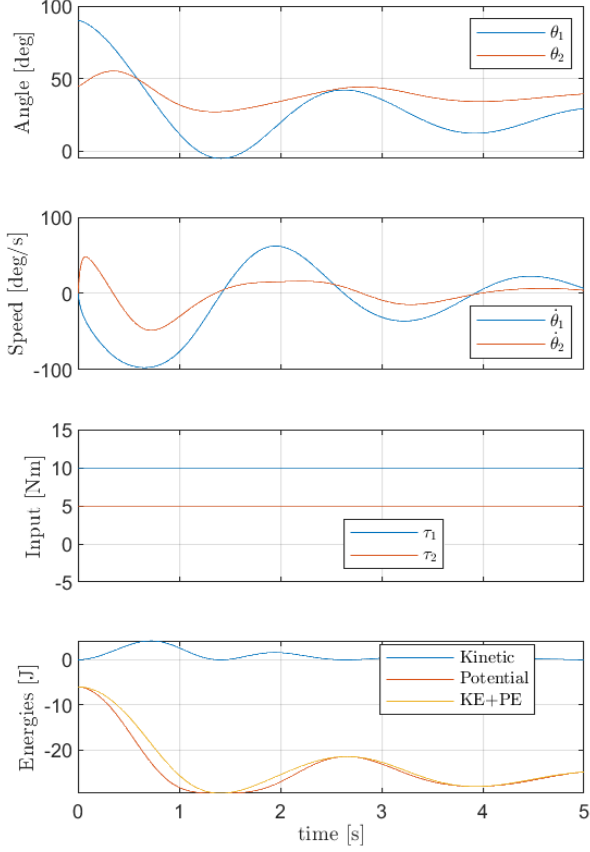


Fig. 3. Results of ODE45 integration for control inputs of 5 and 10. Response is reasonable.

same form as the EOMs (without inverting the M matrix) including the $\dot{x}$ terms as slack variables. Doing so makes it possible to export the symbolic grads and Jacobians from the symbolic math toolbox. Before introducing these slack variables, Matlab's symbolic math toolbox was incapable of exporting the analytic Jacobian for the constraint matrix. These defect constraints and the above task constraints are handled as equality constraints by Matlab's fmincon.

$$0 = -x_{i+1} + x_i + \frac{dt}{2}(\dot{x}_{i+1}(x_{i+1}, \tau_{i+1}) + \dot{x}_i(x_i, \tau_i))$$

B. Design Vector

where

$$x = \begin{pmatrix} \tau_{11} \\ \vdots \\ \tau_{1n} \\ \tau_{21} \\ \vdots \\ \tau_{2n} \\ \theta_{11} \\ \vdots \\ \theta_{1n} \\ \theta_{21} \\ \vdots \\ \theta_{2n} \\ \dot{\theta}_{11} \\ \vdots \\ \dot{\theta}_{1n} \\ \dot{\theta}_{21} \\ \vdots \\ \dot{\theta}_{2n} \\ \ddot{\theta}_{11} \\ \vdots \\ \ddot{\theta}_{1n} \\ \ddot{\theta}_{21} \\ \vdots \\ \ddot{\theta}_{2n} \\ t_f \end{pmatrix}$$

By inspection we see that the design vector includes $8N + 1$ elements, where N is the number of knot points in the direct collocation optimization.

VI. CONTROLLER DESIGN AND SOLVER

Now that we have transcribed the optimization problem we need to decide how many knot points to include. Since I am using trapezoidal integration for my defect constraints the dynamics will be second order accurate with regard to the grid resolution. Validating the input profiles with ODE45 it was determined that 40 knot points were necessary. Using Matlab's fmincon function we discover that these optimizations require approximately 15 - 30 minutes to converge. Supplying the analytical gradients and Jacobian functions for the objective and constraints allows fmincon to converge in approximately 30 seconds.

A. Results

Figure 4 shows the results of our minimum time trajectory optimization for the commanded pose of 60° for both bend angles. Qualitatively speaking it was noticed that optimal trajectories tend to keep the state vectors θ_1 and θ_2 180° out of phase as well as their derivatives $\dot{\theta}_1$ and $\dot{\theta}_2$ while arriving at a specified pose.

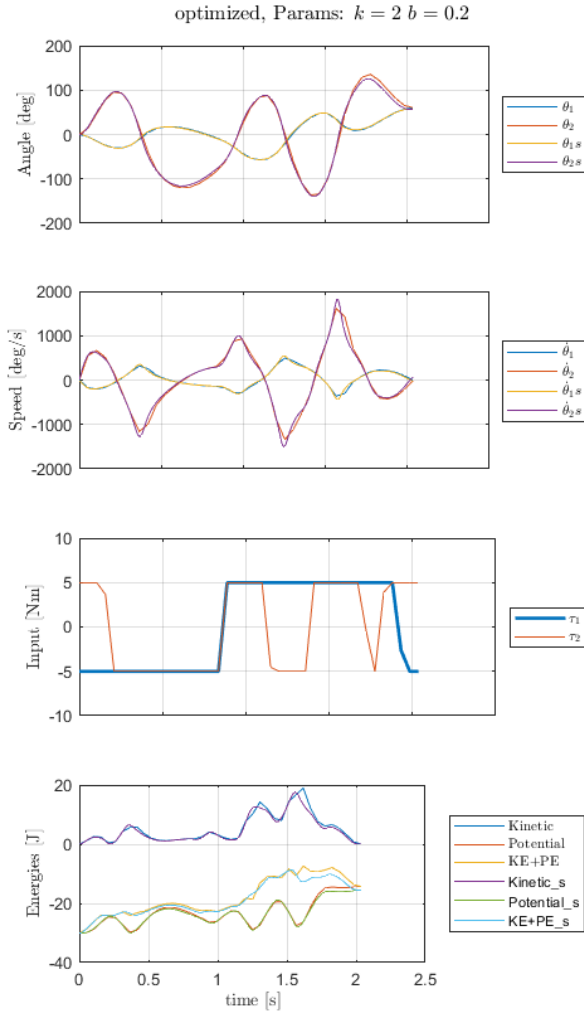


Fig. 4. Example Optimization that discovered the control inputs that minimize time traveling from $0^\circ 0^\circ$ to $60^\circ 60^\circ$. Note the variation between Trapezoidal integration and ODE45 integration of these inputs.

Since the addition of analytical gradients and Jacobians allows for rapid solutions, we go a step further and pose the question of which bend angles in the actuator space can be achieved satisfying our trajectory optimization. Figure 5 shows which bend angle combinations converged and which failed to converge. This plot would be different for each combination of damping and stiffness and maximum and minimum control effort. For this particular system, we find that the top actuator is the performance bottleneck since any pose can be achieved by the bottom actuator, but the top is limited to angles less than 90° .

VII. FUTURE WORK

I am interested in expanding this work to include both the flight phase and the stance phase so that I can model Lamprey inspired climbing. Adding a rotational degree of

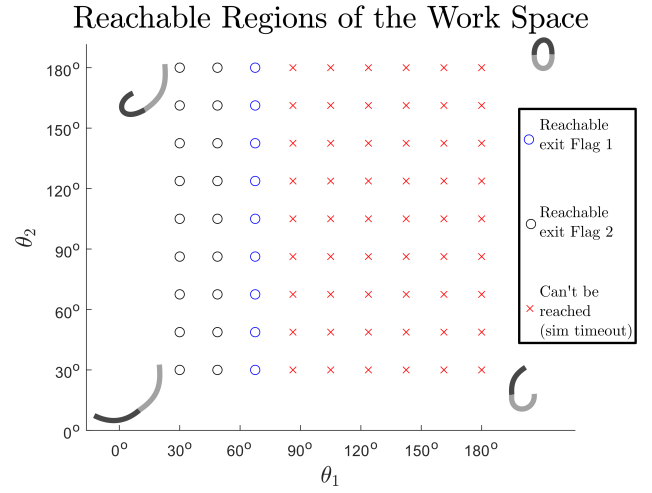


Fig. 5. Results of a nine by nine node parameter sweep of the actuator workspace revealing the poses that are can be reached given the state and torque constraints

freedom to the work here would be sufficient for the stance phase. Adding to that two translation degrees of freedom for the head would give the flight phase. The author would like thank Dr. Christian Hubicki for his instruction in optimal control methods.

REFERENCES

- [1] Danforth, S. M., Kohler, M., Bruder, D., Rabosky, A. R. D., Kota, S., Vasudevan, R., & Moore, T. Y. (2020). Emulating duration and curvature of coral snake anti-predator thrashing behaviors using a soft-robotic platform. arXiv preprint arXiv:2002.09577, (), .
- [2] Tjavaras, A. A. (1996). The dynamics of highly extensible cables (Doctoral dissertation). Massachusetts Institute of Technology, .
- [3] Marchese, A. D., Tedrake, R., & Rus, D. (2016). Dynamics and trajectory optimization for a soft spatial fluidic elastomer manipulator. The International Journal of Robotics Research, 35(8), 1000-1019.
- [4] Althoff, M., Patil, M. J., & Traugott, J. P. (2012). Nonlinear modeling and control design of active helicopter blades. Journal of the American Helicopter Society, 57(1), 1to11.
- [5] Walker, I. D. (2013). Continuous backbone continuum robot manipulators. Isrn robotics, 2013(), .